

MoleBio



Interactive Biochemistry & Molecular Biology

54 modules · 27 viewer types · 5 calculators · Zero install

molebiolab.github.io/MoleBio | ko-fi.com/molebio

Why MoleBio?

Biochemistry is 3D

Flat textbook diagrams can't capture how molecules fold, bind, and interact. MoleBio lets you rotate, zoom, and explore real molecular structures.

Tools are fragmented

Students juggle separate tools for molecular viewing, video lectures, flashcards, and textbook PDFs. MoleBio integrates viewers, animations, calculators, and quizzes in one place.

Learning should be interactive

Step through metabolic pathways reaction by reaction. Adjust enzyme parameters in real time. See how mutations affect protein folding.

What is MoleBio?

MoleBio is an all-in-one interactive learning platform for biochemistry and molecular biology. It runs entirely in your browser — no downloads, no accounts, no ads.

3D Molecular Viewers

Explore real protein structures from RCSB with ball-and-stick, hydrophobicity, and charge rendering modes

Pathway Animations

Step through glycolysis, TCA, ETC, and 10+ metabolic pathways

Real-Time Calculators

Michaelis-Menten curve fitter, DNA T_m, protein MW/pI, and more

Inline Quizzes & Anki Export

Practice questions with spaced repetition scoring per module

Desktop Layout

≡

MoleBio

V1.0



Try the practice!

PROGRESS

0%

0/54 modules

Filter modules...

INTRODUCTION

 **Biochemistry Fundamentals**
10 subtopics

Scale & Macromolecules •

Units & Molarity

Amino Acids

Peptide Bond Formation

Nucleotides & Bases

Base Pairs in Context

DNA & RNA Structure

Micropipettes

Gel Electrophoresis

Western Blot

TOPICS

 **Protein Structure & Folding**
7 subtopics

 **Enzyme Kinetics**
3 subtopics

 **Lipids & Membranes**

BIOCHEMISTRY FUNDAMENTALS

Biological Scale & Macromolecules

Life is built from four major classes of macromolecules: proteins, nucleic acids, carbohydrates, and lipids. These molecules span an enormous range of sizes — from small metabolites measured in angstroms to chromosome-scale DNA molecules meters long. Understanding their scale and basic properties is the foundation of biochemistry.

PROTEINS

Polymers of Amino Acids

Chains of amino acids (typically 50–30,000+ residues) folded into precise 3D structures. They catalyze reactions (enzymes), provide structure (collagen, keratin), signal (hormones, receptors), and transport molecules (hemoglobin).

NUCLEIC ACIDS

Polymers of Nucleotides

DNA stores genetic information as a double helix; RNA transmits and regulates it. The human genome is ~3.2 billion base pairs (~2 meters of DNA per cell), packed into a nucleus ~6 μm across.

CARBOHYDRATES

Sugars & Polysaccharides

From simple sugars (glucose, 180 Da) to massive polysaccharides (glycogen, starch, cellulose). Serve as energy sources, structural components (cell walls), and recognition signals (glycoproteins).

LIPIDS

Hydrophobic & Amphipathic

Not true polymers but essential macromolecules. Fatty acids, phospholipids (membranes), steroids (cholesterol, hormones), and triglycerides (energy storage). Defined by hydrophobicity.

Deep Dive

Biological molecules span roughly 10 orders of magnitude in size. A water molecule is ~2.75 Å across. Glucose is ~7 Å. A typical globular protein like hemoglobin is ~55 Å (5.5 nm). A ribosome is ~250 Å (25 nm). A mitochondrion is ~1,000 nm (1 μm). A typical animal cell is ~10–30 μm . All four macromolecule classes are built from simple monomers linked by condensation (dehydration) reactions: amino acids → proteins, nucleotides → nucleic acids, monosaccharides →

Scale & Macromolecules •

RCSB PDB

Practice

Biological Scale

4 Macromolecule Classes

Biological molecules span ~5 orders of magnitude in size. The bars below use a logarithmic scale.

 Water		2.75 Å
 Glucose		7 Å
 Amino acid		~5 Å
 ATP		~10 Å
 tRNA		~75 Å
 Hemoglobin		5.5 nm
 Ribosome		25 nm
 Virus (HIV)		120 nm
 Mitochondrion		1,000 nm
 E. coli		2,000 nm
 Animal cell		~20 μm

Scale note: This chart uses a logarithmic scale. A ribosome (25 nm) is ~100× wider than water (0.275 nm). An animal cell is ~800× wider than a ribosome. DNA in a single human cell stretches ~2 meters if straightened — packed into a 6 μm nucleus.

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At a Glance

54

Learning
Modules

27

Interactive
Viewer Types

5

Real-Time
Calculators

229

+

Quiz
Questions

- ✓ Zero install — runs in any modern browser
- ✓ Real PDB molecular data from RCSB
- ✓ No accounts, no ads, no tracking cookies
- ✓ Works offline as a Progressive Web App (PWA)
- ✓ Touch-friendly 3D rotation on mobile
- ✓ Anki CSV export for spaced repetition

3D Molecular Viewers

Explore real protein and molecular structures in interactive 3D — directly in your browser.

Real PDB Data

Fetches actual protein structures from the RCSB Protein Data Bank. See hemoglobin, DNA polymerase, ATP Synthase — not simplified cartoons.

Ball-and-Stick Models

Hardcoded atom coordinates for amino acids, nucleotides, and metabolic intermediates with chemically accurate bond angles.

Touch Rotation

Drag to rotate, pinch to zoom on any device. Designed for both desktop mouse interaction and mobile touch.

Amino Acid Gallery

All 20 standard amino acids with 3D models showing side chain properties, charge states, and hydrophobicity.

Interactive Viewers

MoleBio

V1.0



INTRODUCTION

Filter modules...

10 subtopics

TOPICS



Protein Structure & Folding

7 subtopics

Levels of Structure

Secondary Structure Animator

Secondary Structure Builder

Ramachandran Explorer

Folding Mechanisms

Chaperones

Misfolding & Disease



Enzyme Kinetics

3 subtopics



Lipids & Membranes

3 subtopics



Metabolic Pathways

13 subtopics



DNA, RNA & Gene Expression

9 subtopics



Signal Transduction

4 subtopics

PROTEIN STRUCTURE & FOLDING

Molecular Chaperones

Not all proteins fold spontaneously in the crowded cellular environment. Molecular chaperones prevent premature aggregation and provide protected environments where folding can proceed correctly.

HSP70

Holdases

Bind exposed hydrophobic patches on newly synthesized chains, preventing aggregation co-translationally.

GROEL/ES

Foldases

Barrel-shaped cage (Anfinsen cage) providing an isolated hydrophilic environment for a single protein molecule to fold.

HSP90

Conformational Regulators

Stabilize metastable signaling proteins including kinases and hormone receptors. Upregulated in many cancers.

TRIC/CCT

Eukaryotic Chaperonin

Eukaryotic equivalent of GroEL/ES. Essential for folding actin, tubulin, and other cytoskeletal proteins.

Deep Dive

The GroEL/GroES system is the best-characterized chaperonin. GroEL forms a double-ring barrel. Substrates bind in the central cavity. GroES caps the ring, enclosing the substrate in a hydrophilic environment. ATP hydrolysis drives large allosteric conformational changes.

In E. coli, ~10–15% of newly synthesized proteins require GroEL/GroES for proper folding. Obligate clients are enriched in TIM-barrel topologies prone to kinetic trapping.

Key Takeaways

-> Hsp70 (DnaK) binds hydrophobic segments of nascent chains emerging from the ribosome

Ribbon Viewer

RCSB PDB

Practice

1A0N: Crystal Structure of the Asymmetric Chaperonin Com

RCSB

68870 atoms · 3.00 Å · X-ray · Escherichia Coli · 21 chains · 189 ligand atoms

Ribbon

Ball+Stick

Surface

Spin

1A0N

Load PDB

Color: Chain 2° Structure Hydrophobicity Charge

Measure

GroEL apical domain

GroES mobile loop

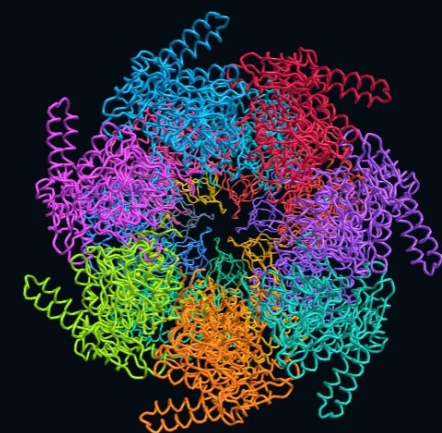
Loaded from RCSB

GroEL-GroES

PDB: 1A0N

Hsp90 dimer

PDB: 4B9Q



click atom for info · drag to rotate · scroll to zoom · arrow keys & +/-

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Calculators & Tools

DNA Tm Calculator

Nearest-neighbor thermodynamic model (SantaLucia 1998). Adjustable salt, primer concentration, and mismatches.

Protein MW & pI

Calculate molecular weight and isoelectric point from amino acid sequence. Uses ExPASy-validated parameters.

MM Curve Fitter

Paste kinetic data, fit Michaelis-Menten curves with Gauss-Newton regression. Extract K_m and V_{max} .

Central Dogma Flow

Visualize the flow from DNA → RNA → Protein interactively. Input a sequence and see transcription and translation.

Sequence Converter

Convert between DNA, RNA, and protein sequences. Complement, reverse complement, and reading frame tools.

Quizzes & Assessment

Every module includes built-in practice questions with instant feedback and spaced repetition scoring.

Per-Module Questions

Targeted practice questions for each topic, from amino acid properties to pathway regulation. 229+ questions across 30 modules.

Spaced Repetition

Built-in scoring tracks which concepts you've mastered and which need review, mimicking proven SRS learning techniques.

Anki CSV Export

Export any module's questions as Anki-compatible CSV files. Integrate MoleBio content into your existing flashcard workflow.

Instant Feedback

See explanations immediately after answering. Understand why the correct answer is right — not just which one it is.

Curated Modules

Biochemistry Fundamentals

10 modules

Amino acids, peptide bonds, nucleotides, base pairing, DNA/RNA, and lab techniques.

Enzyme Kinetics

3 modules

Michaelis-Menten kinetics, inhibition modes, allostery, and cooperative binding.

Metabolic Pathways

13 modules

Glycolysis, TCA, ETC, β -oxidation, urea cycle, nucleotide synthesis, and more.

Signal Transduction

4 modules

GPCR/cAMP, receptor tyrosine kinases, JAK-STAT, and NF- κ B pathways.

Protein Structure & Folding

7 modules

Primary through quaternary structure, Ramachandran plots, and chaperones.

Lipids & Membranes

3 modules

Phospholipids, cholesterol, membrane architecture, and transport mechanisms.

DNA, RNA & Gene Expression

9 modules

Replication, transcription, translation, CRISPR, splicing, and epigenetics.

Calculators & Tools

5 modules

DNA T_m, protein MW/pI, central dogma flow, sequence converter, and MM curve fitter.

Works **Everywhere**

One app. Every device. No compromises.

Desktop

Full 3-column layout with sidebar navigation, lesson panel, and interactive viewer side by side.

Mobile

Responsive hamburger menu, stacked panels, touch-optimized 3D rotation. Designed mobile-first.

Offline / PWA

Install as a Progressive Web App. Full functionality without an internet connection after first load.

Zero Install

No downloads, no plugins, no setup. Just open a URL. Works in Chrome, Safari, Firefox, Edge.

Who It's For

Pre-Med Students

Master biochemistry for MCAT with interactive pathway animations and real molecular data.

Medical & Pharmacy Students

High-yield review of metabolic pathways, drug metabolism, enzyme kinetics, and clinical biochemistry.

Professors & Instructors

Classroom supplement. Project MoleBio during lectures or assign modules as homework.

Graduate Students

Quick-reference viewers and calculators for research, qualifying exams, and teaching prep.

Edtech & University Licensing

Integrate MoleBio modules into your LMS, curriculum, or platform. Custom content partnerships available.

Biochemistry & Molecular Biology Majors

Go deeper than your textbook — explore protein folding, enzyme kinetics, and signaling hands-on.

AP/IB Biology Students

Visual, interactive introduction to molecular biology beyond typical high school resources.

Nursing & PA Programs

Focused biochemistry review without the complexity overhead of research-grade tools.

Biotech & Pharma Professionals

Refresh core biochemistry concepts with real PDB structures and pathway animations.

Science Enthusiasts

Curious about how life works at the molecular level? Explore MoleBio — no background required.

Under the Hood

Architecture

Ships as a single HTML file — 11,300 lines of production code, zero dependencies

Frontend

Vanilla HTML/CSS/JS — no frameworks, no build step, no package manager

3D Rendering

Three.js (deferred load) with WebGL memory leak prevention and cleanup

Data Storage

localStorage for progress, quiz scores, and spaced repetition — no server, no personal data collected

Hosting

GitHub Pages with Progressive Web App (PWA) manifest and service worker for offline support

SEO

Open Graph, Twitter Cards, JSON-LD, sitemap, canonical URLs

Accessibility

ARIA labels, keyboard navigation, focus trapping, reduced-motion support

MoleBio



Built to make biochemistry and molecular biology digestible for everyone.

Try it now

molebiolab.github.io/MoleBio

Support the project

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Every contribution helps keep MoleBio growing — thank you for your support!